

Environmental Toxicology

ACUTE TOXICITY OF SODIUM BICARBONATE, A MAJOR COMPONENT OF COAL BED NATURAL GAS PRODUCED WATERS, TO 13 AQUATIC SPECIES AS DEFINED IN THE LABORATORY

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Abstract: Water produced during coal bed natural gas (CBNG) extraction in the Powder River Structural Basin of Wyoming and Montana (USA) may contain concentrations of sodium bicarbonate (NaHCO_3) of more than 3000 mg/L. The authors evaluated the acute toxicity of NaHCO_3 , also expressed as bicarbonate (HCO_3^-), to 13 aquatic organisms. Of the 13 species tested, 7 had a median lethal concentration (LC_{50}) less than 2000 mg/L NaHCO_3 , or 1300 mg/L HCO_3^- . The most sensitive species were *Ceriodaphnia dubia*, freshwater mussels (*Lampsilis siliquoidea*), pallid sturgeon (*Scaphirhynchus albus*), and shovelnose sturgeon (*Scaphirhynchus platyrhynchus*). The respective LC_{50} s were 989 mg/L, 1120 mg/L, 1249 mg/L, and 1430 mg/L NaHCO_3 , or 699 mg/L, 844 mg/L, 831 mg/L, and 1038 mg/L HCO_3^- . Age affected the sensitivity of fathead minnows, even within life stage. Two days posthatch, fathead minnows were more sensitive to NaHCO_3 and HCO_3^- compared with 4-d-old fish, even though fish up to 14 d old are commonly used for toxicity evaluations. The authors recommend that ion toxicity exposures be conducted with organisms less than 24 h posthatch to ensure that experiments document the most sensitive stage of development. The results of the present study, along with historical and current research regarding the toxicity of bicarbonate, may be useful to establish regulatory standards for HCO_3^- . *Environ Toxicol Chem* 2014;33:525–531. © 2014 SETAC. This article is a US Government work and is in the public domain in the USA.

Keywords: Acute toxicity Sodium bicarbonate Bicarbonate Produced waters Early life stage sensitivity

INTRODUCTION

New energy technologies, such as coal bed natural gas (CBNG) extraction have resulted in a rapid increase in natural gas drilling. These new techniques allow for the exploitation of natural gas reserves that were previously uneconomical. However, to remove gas trapped in coal bed formations, water must be pumped from the coal seam, and the process can result in large quantities of produced or coproduced waters. The chemistry of the water produced is highly variable, depending on the geologic origin of the region. Total dissolved solids (TDS) can range from less than 1000 to more than 400 000 mg/L [1]. The constituents dissolved in the production water are determined by geology and can include sulfate, chloride, bicarbonate, sodium, calcium, potassium, and other elements and compounds. The Tongue and Powder River watersheds of northeastern Wyoming and southeastern Montana (USA) have undergone considerable CBNG development beginning in the late 1990s. Up to 60 000 wells have been permitted, and approximately 844 000 acre-feet of water was produced in the Powder River Structural Basin (which encompasses the Tongue and Powder River watersheds) between 2001 and 2010 [2,3].

While the quality of produced waters is variable within the Powder River Structural Basin, some contain large quantities of sodium bicarbonate [4,5]. Sodium bicarbonate (NaHCO_3) is a generally innocuous substance at low to moderate concentrations and occurs naturally in ground and surface waters. Like all compounds or salts, however, NaHCO_3 exposures to aquatic organisms can result in toxic effects as concentrations approach a critical threshold. Mount et al. [6] found that NaHCO_3 was one

of the most toxic salts that frequently occur in oil field production waters. Hoke [7] and Mount et al. [6] defined the toxic fraction of the salts as the co-occurring anion, which in the case of NaHCO_3 is bicarbonate (HCO_3^-). The sensitivity of aquatic organisms to HCO_3^- is variable. Some fish—such as catfish (Ictaluridae), rainbow trout (Salmonidae), and bluegill (Centrarchidae)—and macroinvertebrates such as Chironomidae, crayfish, and many others are tolerant of HCO_3^- and other salts [8,9]. Other taxa are, however, much more sensitive, including *Daphnia*, fathead minnow, and many mayfly genera [6,10]. The study of Mount et al. [6] was an important first step that experimented with multiple salts, and multiple combinations of salts to predict ion toxicity; however, it was limited to 3 species of aquatic organisms exposed in moderately hard reconstituted water.

The purpose of the present study was to expand the state of scientific knowledge about the acute sensitivity of multiple aquatic species to NaHCO_3 under controlled, laboratory conditions. The ecological significance of the species sensitivity was investigated further because river waters from the basin with NaHCO_3 as the principle component were reconstituted or collected for use as experimental waters. The results of the present study, along with those from an in situ toxicity study, provide a basis for researchers, regulators, and industry to define acceptable discharge of ions to minimize the impact on biota in receiving waters.

MATERIALS AND METHODS

Experimental organisms

To define the toxic threshold of NaHCO_3 and HCO_3^- , acute toxicity experiments were performed with 13 species, 9 of which were native to the Powder and Tongue River drainages: the white sucker (*Catostomus commersoni*), pallid sturgeon (*Scaphirhynchus albus*), shovelnose sturgeon (*Scaphirhynchus*

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platyrhynchus), fathead minnow (*Pimephales promelas*), chironomids (*Chironomus dilutus*), tubifex worms (*Tubifex tubifex*), amphipods (*Hyalella azteca*), freshwater mussels (fatmucket, *Lampsilis siliquoidea*), and *Ceriodaphnia dubia*. Three of the additional species, while not native to the study area, have been introduced as sport fish to the Tongue and Powder River Basins: walleye (*Sander vitreus*), northern pike (*Esox lucius*), and rainbow trout (*Oncorhynchus mykiss*). The African clawed frog (*Xenopus laevis*) was selected as a surrogate amphibian species. Ideally, the native northern leopard frog (*Rana pipiens*), an amphibian native to the Powder and Tongue River Basins, would have been used as an experimental animal. However, because of logistical problems with obtaining eggs and adults at the proper life stage for experiments, and the lack of an approved disease-free source of experimental animals, the clawed frog was used.

Experimental waters

Experiments were conducted in at least 1 of 3 exposure waters: water reconstituted to mimic the water chemistry of the Tongue River during low stream flow when ion concentrations are the greatest (hereafter referred to as Tongue recon); water reconstituted to mimic the water chemistry of the Powder River during low flow when ion concentrations are the greatest (hereafter referred to as Powder recon); and water collected directly from the Yellowstone River near Miles City, Montana (hereafter referred to as Yellowstone). The reconstituted waters were created using a combination of well water from the Jackson Field Research Station, Jackson, Wyoming (USA) and the addition of chemicals of American Chemical Society (ACS) grade or higher or were created at the Montana Fish Wildlife and Parks, Helena, Montana (USA) laboratory using deionized water and chemicals of ACS grade or higher. Additional experiments were conducted with *C. dubia* in moderately hard reconstituted water [11] for direct comparison with previous toxicity investigations.

Exposure protocols

Acute toxicity experiments were conducted according to the protocols described in Bantle et al. [12], ASTM International [13,14], and the US Environmental Protection Agency (USEPA) [11]. The experiments were conducted to define the median lethal concentration (LC50). The duration of toxicity experi-

ments was 96 h for all organisms, except chironomids and *C. dubia*, for which the duration was 48 h. Target or nominal concentration ranges of NaHCO_3 for the exposures were created through the addition of NaHCO_3 to Tongue recon, Powder recon, or Yellowstone control waters for most experiments. Serial dilutions of stock solutions were used to create the exposure concentration ranges for chironomids, amphipods and *Tubifex*.

Aquaria and water quality

Experiments with white sucker, pallid sturgeon, shovelnose sturgeon, fathead minnow, rainbow trout, amphipods, chironomids, *Tubifex*, walleye, and northern pike were conducted in 0.5-L to 4-L acid-washed glass aquaria. African clawed frog exposures were conducted in 10-mL acid-washed Petri dishes. The *C. dubia* experiments were conducted in 30-mL plastic cups. Freshwater mussel experiments were conducted in 30-mL acid-washed glass beakers. All experiments were static renewal, with water replaced every 24 h. Larval or immediate posthatch organisms were used for all experiments (Table 1). The fathead minnow experiment in Tongue recon was initially conducted with fish aged 4 d posthatch (dph). However, the maximum HCO_3^- concentration of approximately 2000 mg/L resulted in insufficient mortality to calculate an LC50. Field studies conducted concurrently led us to believe that younger fish (2 dph vs 4 dph) were more sensitive to HCO_3^- , and an experiment was reinitiated with 2-dph fathead minnows exposed in Tongue recon.

For each experiment, exposure waters containing increasing concentrations of NaHCO_3 were added to create a range of exposures from a control to up to 11 700 mg/L NaHCO_3 (Tables 2 and 3), depending on the sensitivity of the organism determined during range-finding experiments. Dissolved oxygen, temperature ($^{\circ}\text{C}$), and conductivity were measured twice daily. Alkalinity, pH, and hardness were measured 2 to 4 times per experiment. Additional results of measured experimental conditions for each species are described in Farag and Harper [15].

Chemical analysis

Experimental water was collected from serial dilutions 1 to 3 times per experiment. Anion samples were transported in bicarbonate-rinsed 125-mL polyethylene containers and

Table 1. Exposure conditions and experimental organisms used in acute laboratory toxicity experiments with 13 species

Species	Age or life stage	Replicates ^a (No. per replicate)	Exposure water	Exposure range mg/L sodium bicarbonate (mg/L bicarbonate)	Temperature range ($^{\circ}\text{C}$)
White sucker	22 dph	3 ($n = 10$)	Tongue recon and Powder recon	288–6200 (217–4360)	16–17
Pallid sturgeon	4 dph	3 ($n = 10$)	Tongue recon and Powder recon	286–3243 (210–2404)	18–19
Shovelnose sturgeon	4 dph	4 ($n = 5$)	Yellowstone	181–9930 (115–7320)	15–17
Fathead minnow	2 dph and 4 dph	3 ($n = 10$)	Tongue recon and Powder recon	288–3243 (193–2063)	23–26
Rainbow trout	2 dph	3 ($n = 10$)	Tongue recon	387–13820 (248–10800)	10–11
Walleye	1 dph	4 ($n = 5$)	Yellowstone	351–4840 (209–3540)	14–15
<i>Ceriodaphnia dubia</i>	Neonates	10 ($n = 1$)	Tongue recon, Powder recon and Mod-hard recon	114–2475 (93–1707)	24–27
Freshwater mussels	Newly transformed juveniles	4 ($n = 5$)	Tongue recon	256–2488 (211–1870)	19–22
African clawed frog	Embryos	2 ($n = 25$)	Tongue recon	265–3823 (220–2730)	22–24
Chironomids	Larvae (7–14 d)	4 ($n = 10$)	Tongue recon and Powder recon	295–15 910 (190–11 700)	21–23
Tubifex	Adults	4 ($n = 10$)	Tongue recon and Powder recon	319–5060 (207–3460)	17–19
Amphipods	Adults	4 ($n = 10$)	Tongue recon and Powder recon	255–1794 (147–1210)	23
Northern pike	1 dph	4 ($n = 10$)	Yellowstone	157–5284 (109–4690)	10–11

^aNumbers of replicates and organisms per replicate (n) were selected following US Environmental Protection Agency and ASTM International guidelines, unless a limited number of organisms were available.

recon = reconstituted; dph = days posthatch.

Table 2. Mean ion concentrations (mg/L) measured in reconstituted Tongue River exposure water

Sample size	HCO ₃ ⁻	Ca ²⁺	Mg ²⁺	Na ⁺	K ⁺	Cl ⁻	SO ₄ ²⁻	NaHCO ₃ nominal
<i>Ceriodaphnia dubia</i>								
3	249	47	34	43	2.4	1.7	157	Control
3	610	44	34	185	2.0	2.3	164	900
3	715	38	33	303	2.7	2.0	157	1300
3	810	30	33	390	2.7	2.2	162	1600
3	1000	4.6	33	476	3.3	2.8	158	2000
3	1460	5	32	765	4.1	3.9	171	3000
Fatmucket mussel								
2	211	39.0	33	45	2.4	2.2	142	Control
2	390	29	33	131	2.3	2.2	143	500
2	623	22.4	33.2	189	2.16	1.4	152	750
2	810	20.5	33.4	255	2.17	1.1	151	1000
2	1100	16.4	32.6	367	2.11	1.2	150	1500
2	1480	8.8	32	511	2.5	1.3	144	2000
2	1870	14	31	618	2.4	1.2	143	2500
Pallid sturgeon								
3	210	42.3	20.6	75.9	7.41	8.0	186	Control
3	363	40.4	20.5	134	7.39	7.9	186	518
3	543	38.5	20.5	193	7.37	7.8	186	864
3	844	34.6	20.35	309	7.33	7.6	186	1440
3	1360	26.9	20.1	543	7.25	7.1	185	2400
3	2404	11.5	19.6	1,110	7.1	6.3	185	4000
Amphipods								
1	147	2.0	19.6	108	6.03	3.3	177	Control
1	1366	2.5	18.8	551	7.0	2.8	175	2000
Fathead minnow								
3	210	47.0	34	45	2.5	2.2	140	Control
3	374	47	34	134	2.4	2.3	142	500
3	627	37	34	216	2.4	2.2	42	800
3	1020	22	34	390	2.6	2.4	146	1400
3	1330	16	33	524	2.8	4.0	143	2000
3	1970	12	34	812	3.4	5.1	148	3000
African clawed frog								
3	220	45	33	45	2.4	2	139	Control
3	325	33	30	131	2.3	2.2	143	500
3	508	18	30	193	4	3.2	132	750
3	901	13	31	390	2.7	2.7	136	1500
3	1730	16	31	628	2.7	3.5	136	2500
3	2330	4.3	31	901	2.8	4.9	135	3500
3	2730	3.6	31	1,093	2.8	5.4	135	4500

refrigerated. Cation samples were held in 125-mL I-ChemTM containers and preserved at pH < 2.0. All water samples were refrigerated and sent to the Montana Department of Health and Human Services laboratory in Helena, Montana (USA), for analyses. Water samples were analyzed for cations using USEPA method 200.7; Determination of metals and trace elements in water and wastes by inductively coupled plasma-atomic emission spectrometry [16]. Spike recovery ranged from 91.4% to 102.2%, and relative differences between duplicate samples were less than 5%. Anion samples were measured using USEPA Method 300.0; Determination of inorganic anions by ion chromatography [17]. Spike recovery ranged from 98% to 107%.

Statistical analysis

Median lethal concentrations were calculated with the Toxicity Relationship Analysis Program (TRAP) [18] using measured NaHCO₃ and HCO₃⁻ values. The predicted toxicity of salt combinations in experimental waters was tested on fathead minnows and *C. dubia* using the salinity toxicity relationship (STR) model [19].

RESULTS

Median lethal concentrations

Seven species had acute LC50 concentrations under 2000 mg/L NaHCO₃ (1300 mg/L HCO₃⁻). The most sensitive

species were *C. dubia*, fatmuckets, pallid sturgeon, shovelnose sturgeon, amphipods, fathead minnows, and African clawed frogs. Five of the species had LC50s under 1500 mg/L NaHCO₃ (Table 4 and Figure 1). The NaHCO₃ LC50 for *C. dubia* was 989 mg/L with a 95% confidence interval (CI) of 910 mg/L to 1069 mg/L in Tongue recon, 1355 mg/L (95% CI = 1257–1453 mg/L) in Powder recon, and 1288 mg/L (95% CI = 1171–1406) in moderately hard reconstituted water. The LC50 for fatmuckets in Tongue recon was 1120 mg/L NaHCO₃ (95% CI = 1099–1497 mg/L). The LC50 for pallid sturgeon was 1295 mg/L NaHCO₃ (95% CI = 927–1662 mg/L) in Tongue recon and 1356 mg/L (95% CI = 1206–1507 mg/L) in Powder recon.

The shovelnose sturgeon LC50 in Yellowstone River water was 1430 mg/L NaHCO₃; however, because of inadequate partial effects, accurate confidence intervals could not be calculated. The NaHCO₃ LC50 for fathead minnows was 1749 mg/L in Tongue recon (95% CI = 1604–1895 mg/L) and 2078 mg/L (95% CI = 769–3388 mg/L) in Powder recon.

The amphipod bicarbonate LC50 was 1419 mg/L (95% CI = 1340–1492 mg/L) in Tongue recon. However, low chloride concentrations in the exposure water likely affected survival of amphipods in Tongue recon, where chloride concentrations ranged from 2 mg/L to 2.5 mg/L. Survival of amphipods was near 100% in Powder recon, where chloride ranged from 97.5 mg/L to 103 mg/L. An LC50 for amphipods in Powder

Table 3. Mean ion concentrations (mg/L) measured in exposure waters

Sample size	HCO ₃ ⁻	Ca ²⁺	Mg ²⁺	Na ⁺	K ⁺	Cl ⁻	SO ₄ ²⁻	Species	NaHCO ₃ nominal
Powder River reconstituted water									
2	239	13	180	283	13	11	1320	CDUB	Control
2	505	11	174	438	15	11	1430	CDUB	900
2	661	12	177	488	15	11	1410	CDUB	1300
2	776	12	176	583	16	10	1410	CDUB	1600
2	944	6	176	740	16	10	1330	CDUB	2000
2	1370	6	176	931	17	12	1530	CDUB	3000
3	252	68.9	87.9	324	16.8	98.6	856	PAL	Control
3	535	61.5	87.5	429	16.8	98.8	855	PAL	864
3	887	54	87.1	533	16.7	98.9	853	PAL	1440
3	1470	39.1	86.4	742	16.7	99.3	851	PAL	2400
3	2200	9.4	84.9	1160	16.5	100	846	PAL	4000
1	234	55.4	90.1	288	13	97.5	829	AMP	Control
1	1305	78.9	88	722	13.2	105	885	AMP	2000
3	193	78.3	92.3	285	11.8	101	929	FHM	Control
3	304	76.3	92	332	12	100.4	923	FHM	518
3	751	67	91	557	13.1	97.4	892	FHM	1440
3	1223	57.2	89.5	791	14.2	94.3	861	FHM	2400
3	2063	40.9	87.3	1180	16.1	89.1	808	FHM	4000
Moderately hard reconstituted water									
2	93	15	9	21	7.9	1.4	96	CDUB	Control
2	579	14	13	216	2.7	1.8	99	CDUB	900
2	727	8	13	339	2.4	1.0	93	CDUB	1300
2	944	5	12	418	2.2	1.0	98	CDUB	1600
2	1034	4.0	13	476	2.2	1.2	96	CDUB	2000
2	1707	5.4	34	768	6.9	3.0	113	CDUB	3000
Yellowstone River water									
3	115	3.75	10.75	66	3.8	3.2	110	SHNS	Control
3	565	3.6	10.7	225	4.2	3	110	SHNS	625
3	1020	3.4	10.6	384	4.5	2.9	110	SHNS	1250
3	1920	3	10.5	710	5.3	2.5	110	SHNS	2500
3	3720	1.9	10.4	1,338	6.7	1.9	110	SHNS	5000
3	7320	0.5	10.1	2,610	9.7	0.5	110	SHNS	10 000

CDUB = *Ceriodaphnia dubia*; PAL = pallid sturgeon; AMP = amphipods; FHM = fathead minnow; SHNS = shovelnose sturgeon.

recon could not be calculated because of 93% survival at the highest NaHCO₃ exposure concentrations (1850 mg/L).

The NaHCO₃ LC50 for the African clawed frog in Tongue recon was 1940 mg/L (95% CI = 1482–2890 mg/L). The remaining species all had NaHCO₃ LC50 values greater than 3200 mg/L, with the most tolerant species, the northern pike, unaffected at more than 8000 mg/L NaHCO₃ (Table 4).

The relative sensitivity of experimental organisms expressed as HCO₃⁻ or alkalinity was similar but not identical to NaHCO₃

(Table 4). For example, the LC50 for alkalinity was lower for mussels (670 mg/L) compared with *C. dubia* (734 mg/L), whereas *C. dubia* was the most sensitive species for both NaHCO₃ (989 vs 1120 mg/L), and HCO₃⁻ (699 vs 844 mg/L).

STR model

The STR model predicted a 96-h LC50 of 1590 mg NaHCO₃/L for fathead minnow experiments in reconstituted Tongue River water. The STR model predicted a 48-h LC50 of 1571 mg

Table 4. Median lethal concentration (LC50) of experimental organisms (mg/L)^a

Species	Tongue River (reconstituted)			Powder River (reconstituted)			Yellowstone River ^b		
	Alk	HCO ₃ ⁻	NaHCO ₃	Alk	HCO ₃ ⁻	NaHCO ₃	Alk	HCO ₃ ⁻	NaHCO ₃
<i>Ceriodaphnia dubia</i>	734	699	989	883	769	1,355	846 ^c	842 ^c	1288 ^c
Fatmucket	670	844	1120	—	—	—	—	—	—
Pallid sturgeon	846	939	1295	608	831	1356	—	—	—
Shovelnose sturgeon	—	—	—	—	—	—	880	1038	1430
Amphipod	966	1085	1419	1110 ^d	1037 ^d	2000 ^d	—	—	—
Fathead minnow	1139	1254	1749	1089	1418	2078	—	—	—
African clawed frog	856	914	1940	—	—	—	—	—	—
Walleye	—	—	—	—	—	—	2022	2352	3249
Tubifex	2119	2455	3297	2427	2532	3367	—	—	—
White sucker	2668	3208	4494	3852	4276	5648	—	—	—
Chironomid ^c	2903	3593	4947	3874	5820	8014	—	—	—
Rainbow trout	5134	5699	8070	—	—	—	—	—	—
Northern pike	—	—	—	—	—	—	4181 ^d	8000 ^d	8000 ^d

^aAll exposures were 96 h, except for *C. dubia* and Chironomids, which were 48 h.

^bYellowstone River water collected near Miles City, Montana, USA.

^cUS Environmental Protection Agency moderately hard reconstituted water.

^dLC50 maximum exposure concentrations were not great enough to accurately calculate LC50.

ALK = alkalinity (mg/L as CaCO₃).

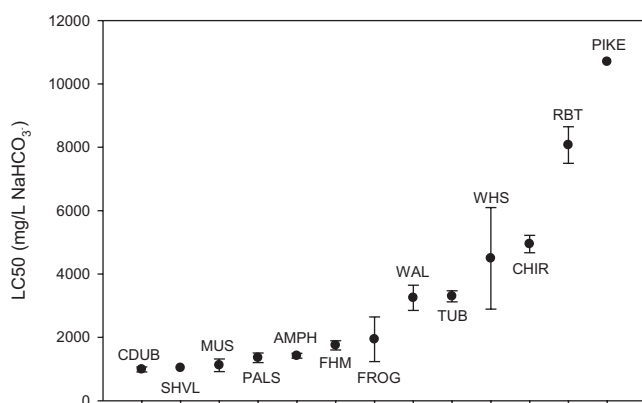


Figure 1. Sodium bicarbonate (NaHCO_3^-) mean lethal concentrations (LC50s) and 95% confidence intervals. CDUB = *Ceriodaphnia dubia*; MUS = fatmucket mussel; FROG = African clawed frog; PALS = pallid sturgeon; SHVL = shovelnose sturgeon; AMPH = Amphipods; FHM = fathead minnow; WAL = walleye; TUB = tubifex; WHS = white sucker; CHIR = Chironomidae; RBT = rainbow trout; Pike = northern pike.

NaHCO_3/L for *C. dubia* in moderately hard reconstituted water; 1776 mg NaHCO_3/L in reconstituted Tongue River water; and 1575 mg NaHCO_3/L in reconstituted Powder River water.

Water chemistry

The mean concentration of ions measured in water collected from experiments with the 7 most sensitive species varied slightly as a result of interlaboratory differences and water preparations (Tables 2 and 3; additional measurements for other experimental species may be found in Farag and Harper [15]). The calcium concentrations in all experiments, measured in waters following serial renewal replacement, decreased as exposure concentrations of sodium bicarbonate increased. Calcium precipitation onto the exposure aquaria was visible, especially at the higher concentrations ($>2000 \text{ mg/L NaHCO}_3$).

DISCUSSION

Sensitivity to NaHCO_3

The acute toxicity of sodium bicarbonate (or bicarbonate) is variable among organisms and is affected by relatively slight differences in the age of the experimental organism. Among the commonly tested laboratory organisms used during the present study, *C. dubia* was the most sensitive. Mount et al. [6] found that *C. dubia* were sensitive to NaHCO_3 with an LC50 of 1020 mg/L. Experiments in the present study with *C. dubia* resulted in an LC50 of 989 mg NaHCO_3/L in simulated Tongue River water, and 1355 mg NaHCO_3/L in simulated Powder River water. Mount et al. [6] also exposed fathead minnow to NaHCO_3 , and found them to be more sensitive than *C. dubia* in moderately hard reconstituted water, with an LC50 of $<850 \text{ mg/L}$. Fathead minnows in the present study exposed to NaHCO_3 in Tongue and Powder River reconstituted waters were not as sensitive as *C. dubia*, with LC50s of 1749 mg NaHCO_3/L and 2079 mg NaHCO_3/L , respectively. These results suggest that ions such as Mg^{2++} and Ca^{2++} present in the Tongue and Powder River reconstituted waters protected fathead minnows from the effects of NaHCO_3 . In fact, the STR model predicted a 96-h LC50 of 1590 mg NaHCO_3/L for fathead minnow in reconstituted Tongue River water, which is more comparable to the measured 96-h LC50 of 1749 mg NaHCO_3/L in the present study. Although the fathead minnow LC50s were less than those reported in moderately hard reconstituted water, these acute

LC50 concentrations are still well below what is present in produced waters from some regions of the Powder River Structural Basin.

Predicted values in STR model for *C. dubia* were all greater than the LC50 values calculated with the TRAP model in the present study (1288 mg NaHCO_3/L in moderately hard reconstituted water, 989 mg NaHCO_3/L in reconstituted Tongue River water, and 1355 mg NaHCO_3/L in reconstituted Powder River water). The STR model predicted a 96-h LC50 of 1590 mg NaHCO_3/L for fathead minnow in reconstituted Tongue River water, which is comparable to our measured 96-h LC50 of 1749 mg NaHCO_3/L . The STR model was conservative for fathead minnow in reconstituted Tongue River water; however, the STR model underpredicted toxicity to *C. dubia* in all reconstituted waters.

The suitability of using *X. laevis* as a surrogate species for *R. pipiens* is unknown. We found the *X. laevis* to be the fifth most sensitive species tested for sodium bicarbonate (LC50 of 1940 mg/L) and the fourth most sensitive for alkalinity (LC50 of 914 mg/L). However, research into the toxicity of salts to frogs in the genus *Rana* indicates they are relatively tolerant to Ringer's solution and road salts $>3000 \text{ mg/L}$ [20,21]. Because *R. pipiens* are a species of special concern, additional research into the effects of NaHCO_3 is suggested.

There has been some criticism that organisms typically used in laboratory toxicity experiments are not sensitive enough to the effects of salts. However, streams draining the Powder River Structural Basin naturally contain relatively high concentrations of TDS with concurrently elevated conductivities [5]. In a study to monitor the long-term effects of produced water on the middle Powder River in Montana, Stagliano [22] found the macroinvertebrate community to be "non-impaired." That study [22] found a total of 64 taxa, and an average of 28 taxa per site. Mayflies were commonly found at all sites; however, species from the family Ephemerillidae were entirely absent, and species from the family Heptageniidae were very uncommon (only 5 individuals found in all samples from all sites). These families are among the most sensitive to elevated TDS and conductivity. Cormier et al. [10] found that in low-conductivity Appalachian Mountain streams, mayfly abundance was a good predictor of water quality. As conductivities in receiving waters approached $300 \mu\text{S/m}$, most mayflies disappeared from the macroinvertebrate assemblage. In the middle Powder River of Montana, near the Wyoming state line, conductivity during low-flow summer discharge is greater than $1000 \mu\text{S/m}$ [22]. Even so, mayflies are abundant in the middle Powder River, likely because TDS and conductivity are naturally high, and native macroinvertebrates have evolved under these conditions. Therefore, while extremely salt-sensitive experimental organisms are applicable for very soft waters, they may not be suitable for the naturally saline, hard waters of the Powder River Structural Basin and similar waters in the arid western United States and elsewhere.

Mode of toxicity

Mount et al. [6] demonstrated that the toxicity of sodium and calcium salts was the result of the co-occurring anions (specifically Cl^- , sulfate, and HCO_3^-). In the Tongue and Powder River waters that were simulated in the present experiments, HCO_3^- was the predominant co-occurring anion. Therefore, it is likely that the primary source of toxicity of NaHCO_3 can be attributed to HCO_3^- . The probable mode of toxicity from HCO_3^- exposure is through disruption of the mechanisms responsible for ionic regulation. A critical mechanism of ionoregulation in aquatic organisms is through ionocytes

(also known as chloride cells or mitochondria-rich cells). Located within the gills and the epithelia of fish, invertebrates, and amphibians, ionocytes are densely packed with mitochondria. Ionocytes use adenosine triphosphatase (ATP), to move ions across the cell membrane [23–25]. An enzyme involved in this process, sodium–potassium adenosine triphosphatase, (Na^+/K^+ ATPase), also known as the Na^+/K^+ pump, is likely affected by exposure to HCO_3^- . Through this process, ionocytes excrete HCO_3^- in exchange for Cl^- . The disruption to chloride movement at the ionocytes is thought to be the ultimate cause of toxicity. Experiments conducted by Hoke and others [7] to determine the toxicity of alkalinity to *C. dubia* concluded that HCO_3^- was toxic to aquatic organisms when concentrations were sufficiently large enough to interfere with this exchange of Cl^- . However, the specific mode of toxicity from HCO_3^- is not well understood and deserves additional investigation.

The influence of calcium

We observed that Ca^{2+} concentrations in exposure waters declined as NaHCO_3 was added. This has been observed in other research [26], and precipitation of calcium was clearly visible on the aquaria in the present study and was more obvious as concentrations increased. Lasier et al. [26] argued that when toxicant-ameliorating ions, such as Ca^{2+} , precipitate out of solution, protection against divalent contaminants is reduced. Elphick et al. [24] also documented reduced toxicity of salts with increasing hardness. We observed that Ca^{2+} declined with increasing NaHCO_3 concentrations in the present study. However, at the HCO_3^- LC50 exposure concentrations of the most sensitive species (699–1245 mg/L HCO_3^-), Ca^{2+} concentrations generally remained above 10 mg/L to 30 mg/L. In the present study, when Ca^{2+} is combined with Mg^{2+} , the remaining total hardness should remain sufficient to buffer effects for all experiments other than those for *C. dubia* conducted in moderately hard reconstituted water. Furthermore, and most importantly, the effects of calcium precipitation on toxicity should be considered in the context of how organisms in the field, under real world conditions, are affected. If calcium precipitation is affecting toxicity, this will undoubtedly occur in the field. Attempting to mitigate against precipitation of calcium in the laboratory may only create an artificially buffered environment that does not accurately indicate what occurs in streams or receiving waters.

Age effects

Young organisms are often more sensitive than adults; however, the effects of toxicants on different species and life stages of fish are variable [12,27,28]. In the present study, 2-dph and 4-dph fathead minnow and 22-dph and 69-dph white suckers were exposed to a range of HCO_3^- concentrations. In both experiments, the LC50s for younger fish and older fish, respectively, were 1254 mg HCO_3^-/L and approximately 2500 mg HCO_3^-/L for fathead minnow, and 3208 mg HCO_3^-/L and approximately 4360 mg HCO_3^-/L for white suckers. The USEPA [11] guidelines require the use of the most sensitive life stage for toxicity testing, and fathead minnow must be 1-dph to 14-dph. Because fathead minnow older than 4 dph were less sensitive than 2-dph minnows, however, experiments that use even slightly older fathead minnows may miss the most sensitive period of development for these fish. Therefore, it appears that with HCO_3^- toxicity, fish must be exposed immediately posthatch to ensure that testing occurs during the most sensitive developmental period. Also, the effect of age on sensitivity was observed during field experiments in the present

study completed in the Powder River Basin [15]. In these experiments, a statistically significant difference in survival was observed between 2-dph and 6-dph fish, which further supports observations that fish at very early life stages (2–4 dph) are more sensitive than fish even just a few days older.

Effect of chloride

The sodium bicarbonate LC50 for amphipods tested in Tongue recon was 1547 mg/L. This is similar to the concentrations found by Lasier et al [26] and seems to indicate that amphipods are among the more sensitive organisms to bicarbonate toxicity. However, the LC50s are likely influenced by the low chloride concentrations in the experimental waters. Amphipod growth is greater in water enriched with chloride (>10 mg/L); and when chloride concentrations are <10 mg/L, amphipods may be more sensitive to acute toxicity (R. Hockett, USEPA, Duluth, MN, personal communication). The Tongue recon experimental water and the moderately hard reconstituted water used by Lasier et al. [26] contained less than 5 mg/L Cl^- and may have affected the health and susceptibility of amphipods to bicarbonate. A more accurate indication of the lack of sensitivity to bicarbonate may come from amphipods exposed to bicarbonate in Powder recon, which contained approximately 100 mg/L Cl^- . Under these experimental conditions, the amphipods had nearly 100% survival in water that contained more than 1850 mg/L bicarbonate, which is above the LC50 value for the Tongue recon containing small concentrations of Cl^- .

CONCLUSIONS

Acute toxicity from NaHCO_3 or HCO_3^- was observed at concentrations well below those documented in produced waters from CBNG wells in the Powder River Structural Basin. Among common organisms used in laboratory studies, *C. dubia* were the most sensitive to NaHCO_3 and HCO_3^- . In the reconstituted waters used for the present study, the STR model underpredicted toxicity to *C. dubia* but was conservative for fathead minnow. The sensitivity of fathead minnow to HCO_3^- and NaHCO_3 was greatly influenced by age, even within the life stage considered acceptable by the USEPA for toxicity testing [11]. Fish that were 4 dph were much more tolerant of NaHCO_3 than 2-dph fish. Therefore, acute toxicity exposures to salts should be conducted with the youngest possible organisms to ensure that experiments document the most sensitive stage of development. Finally, additional research into the specific mode of toxicity from HCO_3^- deserves additional investigation, as does the sensitivity of the northern leopard frog to NaHCO_3 exposure.

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